Stony Brook Institute at Anhui University

安徽大学纽约石溪学院

Chapter 2 Random variable and their probability distributions

Section 2.5 Marginal distribution

- Marginal distribution function
- The marginal distribution of discrete $r.v.$
- The marginal distribution of continuous $r.v.$

Marginal distribution function

Question: If we know the joint CDF of v.r. X and Y, how can we know the CDF of r. v. X and Y?

$$F(x,y)=P\{X\le x,Y\le y\},\ F(x)=P\{X\le x\},$$

$$P\{X\le x\}=P\{X\le x,Y<\infty\}=F(x,\infty)=F_X(x)$$

marginal PDF of X

Marginal distribution

Definition Let $F(x, y)$ be a joint cdf of a two-dimensional r.v. (X, Y), $F_X(x)$ and $F_Y(y)$ are the cumulative distribution functions of X and Y respectively, then $F_X(x)$ and $F_Y(y)$ are called marginal cumulative distribution functions of X and Y, respectively.

By definition of distribution function, we easily get

$$F_X(x) = P\{X \leq x\} = P\{X \leq x, Y < +\infty\}$$

$$= F(x, +\infty) = \lim_{y \to +\infty} F(x, y)$$

and

$$F_Y(y) = P\{Y \leq y\} = P\{X < +\infty, Y \leq y\}$$

$$= F(+\infty, y) = \lim_{x \to +\infty} F(x, y)$$

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The marginal distribution of discrete $r.v.$

Assume (X, Y) is discrete two-dimensional r.v., and its joint probability distribution is $P\{X = x_i, Y = y_j\} = p_{ij}$. The probability distribution of X is

$$p_{i\cdot} = P\{X = x_i\} = P\{\cup_{j=1}^{\infty}(X = x_i, Y = y_j)\}$$

$$= \sum_{j=1}^{\infty} P\{X = x_i, Y = y_j\} = \sum_{j=1}^{\infty} p_{ij}$$

and the probability distribution of Y is

$$p_{\cdot j} = PY = y_j = P\{\cup_{i=1}^{\infty}(X = x_i, Y = y_j)\}$$

$$= \sum_{i=1}^{\infty} P\{X = x_i, Y = y_j\} = \sum_{i=1}^{\infty} p_{ij}$$

We call $p_{i\cdot}$ and $p_{\cdot j}$ be marginal probability distributions of X and Y respect to (X, Y), respectively.

The marginal distribution of discrete $r.v.$

$Y \backslash X$	x_1	x_2	$\cdots$	x_i	$\cdots$
y_1	p_{11}	p_{21}	$\cdots$	p_{i1}	$\cdots$
y_2	p_{12}	p_{22}	$\cdots$	p_{i2}	$\cdots$
$\vdots$	$\vdots$	$\vdots$		$\vdots$	
y_j	p_{1j}	p_{2j}	$\cdots$	p_{ij}	$\cdots$
$\vdots$	$\vdots$	$\vdots$		$\vdots$	

$$P\{X = x_i\} = \sum_{j=1}^{\infty} p_{ij},\ i = 1,2,\cdots;$$

$$F_X(x) = F(x,\infty) = \sum_{x_i \le x} \sum_{j=1}^{\infty} p_{ij},$$

$$P\{Y = y_j\} = \sum_{i=1}^{\infty} p_{ij},\ j = 1,2,\cdots.$$

$$F_Y(y) = F(\infty, y) = \sum_{y_j \le y} \sum_{i=1}^{\infty} p_{ij}.$$

The marginal distribution of discrete $r.v.$

Example 2.5.1: Calculate the marginal CDF of following joint probability distribution.

$Y \backslash X$	0	1	$p_{\bullet j} = P\{Y = y_j\}$
0	$\frac{12}{42}$ +	$\frac{12}{42}$ +	$\frac{4}{7}$
1	$\frac{12}{42}$ +	$\frac{6}{42}$ +	$\frac{3}{7}$
$p_{i\bullet} = P\{X = x_i\}$	$\frac{4}{7}$	$\frac{3}{7}$	1

The marginal distribution of discrete $r.v.$

Example 1. Let the random variable X take a moderate value among the four integers 1, 2, 3 and 4; Another random variable Y may take an integer in 1 ~ X. Find the $P_{i.}$ and $P_{.j}$ of (X,Y).

Solution: the joint cdf of (x, y) should be obtained first;

(1)

X \ Y	1	2	3	4	$P_{i.}$
1	$\frac{1}{4}$	0	0	0	$\frac{1}{4}$
2	$\frac{1}{8}$	$\frac{1}{8}$	0	0	$\frac{1}{4}$
3	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{12}$	0	$\frac{1}{4}$
4	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{4}$
$P_{.j}$	$\frac{25}{48}$	$\frac{13}{48}$	$\frac{7}{48}$	$\frac{3}{48}$	

The marginal distribution of discrete $r.v.$

(2) The marginal distribution of (X,Y) :

X	1	2	3	4
P_k	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$

Y	1	2	3	4
P_k	$\frac{25}{48}$	$\frac{13}{48}$	$\frac{7}{48}$	$\frac{3}{48}$

The marginal distribution of discrete $r.v.$

Example2: Toss an even coin three times. Let X be the number of head (H) occurrences in the three tosses, and Y be the absolute value of the difference between the number of head occurrences and the number of tail (T) occurrences.

Q: Find the joint and the marginal distribution of (X,Y).

Solution：（X, Y）can take values in: (0,3), (1,1), (2,1), (3,3).

$$P(X=0, Y=3) = (1/2)^3 = 1/8$$

$$P(X=1, Y=1) = 3\cdot(1/2)^3 = 3/8$$

$$P(X=2, Y=1) = 3/8$$

$$P(X=3, Y=3) = 1/8$$

X \ Y	1	3
0	0	1/8
1	3/8	0
2	3/8	0
3	0	1/8

The marginal distribution of discrete $r.v.$

X \ Y	1	3
0	0	1/8
1	3/8	0
2	3/8	0
3	0	1/8

x_i	0	1	2	3
p_i	1/8	3/8	3/8	1/8

y_i	1	3
p_i	6/8	2/8

X \ Y	1	3	$P(X=x_i)$
0	0	1/8	1/8
1	3/8	0	3/8
2	3/8	0	3/8
3	0	1/8	1/8
$P(Y=y_i)$	6/8	2/8	1

It is customary to write the marginal distribution at the edge of the joint distribution table, which is the origin of the marginal distribution.

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The marginal distribution of continuous $r.v.$

If (X, Y) is a continuous two-dimensional r.v., and its joint PDF is f(x,y). So we have

$$F_X(x) = P(X \leq x) = P\{X \leq x, Y \in (-\infty, \infty)\}$$

$$= \int_{-\infty}^{x} \int_{-\infty}^{\infty} f(x, y) dy dx$$

For convenience, we can change variables, by definition of PDF, $f_X(x) = \int_{-\infty}^{+\infty} f(x, y) dy$ is just PDF of X. By the similar argument, $f_Y(y) = \int_{-\infty}^{+\infty} f(x, y) dx$ is the PDF of Y. We call $f_X(x)$ and $f_Y(y)$ as above marginal PDF of X and Y respect to (X, Y), respectively.

The marginal distribution of discrete $r.v.$

Example. Let (X, $\boldsymbol{Y}$) be evenly distributed in the region D surrounded by the straight line $x + y/2 = 1$, x axis and y axis; Calculate the joint pdf and the marginal pdf of ($\boldsymbol{X}$, $\boldsymbol{Y}$).

Solution:

(1) because ($\boldsymbol{X}$, $\boldsymbol{Y}$) obeys uniform distribution, so its pdf is:

$$f(x,y)=\begin{cases}\dfrac{1}{A} & (x,y)\in D\\ 0 & \text{others}\end{cases}$$

The marginal distribution of discrete $r.v.$

(2). The marginal pdf is:

$$f_X(x) = \int_{-\infty}^{+\infty} f(x, y)\, dy$$

when $x \le 0$ or $x \ge 1$ $\quad f(x, y) = 0 \quad$ then $\quad f_X(x) = 0$

when $0 < x < 1$ $\quad f_X(x) = \int_{-\infty}^{+\infty} 1 \cdot dy = \int_0^{2(1-x)} dy = 2(1-x)$

Thus,

$$f_X(x) = \begin{cases} 2(1-x) & 0 < x < 1 \\ 0 & \text{others} \end{cases}$$

Similarly,

$$f_Y(y) = \begin{cases} 1 - \frac{1}{2} y & 0 < y < 2 \\ 0 & \text{others} \end{cases}$$

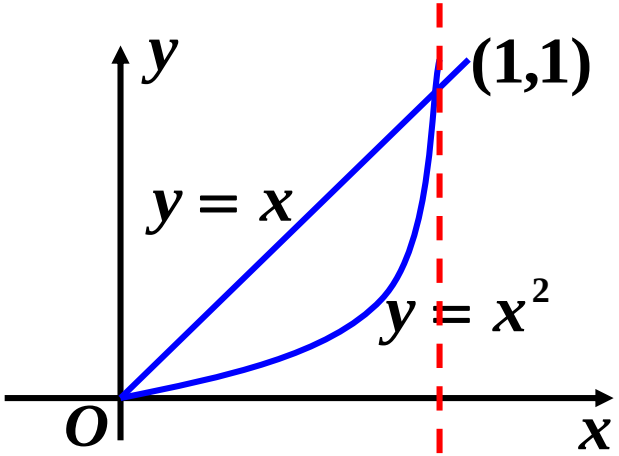
The marginal distribution of continuous $r.v.$

Example 2.5.2: The joint PDF of r.v. X and Y is given by

$$f(x,y)=\begin{cases}6\,, x^2<y<x\\ 0,\ others\end{cases},$$

Calculate the marginal PDF.

Solution:



When $0 \le x \le 1$,

$$f_X(x)=\int_{-\infty}^{\infty} f(x,y)dy \quad =\int_{x^2}^{x} \mathbf{6\,d\,y}$$

$$=\mathbf{6(x-x^2).}$$

When $x<0\ or\ x>1$,

$$f_X(x)=\int_{-\infty}^{\infty} f(x,y)dy=0.$$

$$f_X(\mathbf{x})=\begin{cases}\mathbf{6(x-x^2),} & \mathbf{0\le x\le 1,}\\ \mathbf{0,} & \mathbf{else\ \ others}\end{cases}$$

The marginal distribution of continuous $r.v.$

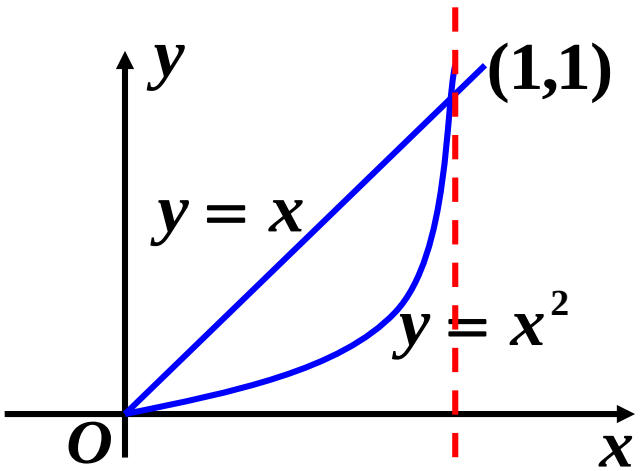
$$f(x,y)=\begin{cases}6\,, x^2<y<x\\ 0, \quad others\end{cases}$$

When $0 \le y \le 1$,

$$f_Y(y)=\int_{-\infty}^{\infty} f(x,y)dx$$
$$=\int_{y}^{\sqrt{y}} 6\,d\,x = 6(\sqrt{y}-y)$$
$$=6(\sqrt{y}-y).$$

When $y<0$ or $y>1$,

$$f_Y(y)=\int_{-\infty}^{\infty} f(x,y)dx = 0.$$



$$f_Y(y)=\begin{cases}6(\sqrt{y}-y), & 0\le y\le 1,\\ 0, & \text{else} \quad others\end{cases}$$

The marginal distribution of continuous $r.v.$

Example 2.5.3: Calculate the marginal PDF of $(X,Y)\sim N(\mu_1,\mu_2,{\sigma_1}^2,{\sigma_2}^2,\rho)$

$$f(x,y)=\frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}}e^{\frac{-1}{2(1-\rho^2)}[\frac{(x-\mu_1)^2}{\sigma_1^2}-\frac{2\rho(x-\mu_1)(y-\mu_2)}{\sigma_1\sigma_2}+\frac{(y-\mu_2)^2}{\sigma_2^2}]}$$

Solution: $\mathbf{f_X(x)} = \int_{-\infty}^{\infty} \mathbf{f(x,y)dy},$

Due to :

$$[\frac{(y-\mu_2)^2}{{\sigma_2}^2}-2\rho\frac{(x-\mu_1)(y-\mu_2)}{\sigma_1\sigma_2}]$$

$$=[\frac{(y-\mu_2)}{\sigma_2}-\rho\frac{(x-\mu_1)}{\sigma_1}]^2-[\rho^2\frac{(x-\mu_1)^2}{{\sigma_1}^2}]$$

The marginal distribution of continuous $r.v.$

$$f_X(x)=\frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}}e^{-\frac{(x-\mu_1)^2}{2\sigma_1^2}}\int_{-\infty}^{+\infty}e^{-\frac{1}{2(1-\rho^2)}\left(\frac{y-\mu_2}{\sigma_2}-\rho\frac{x-\mu_1}{\sigma_1}\right)^2}dy$$

let: $t=\frac{1}{\sqrt{1-\rho^2}}\left(\frac{y-\mu_2}{\sigma_2}-\rho\frac{x-\mu_1}{\sigma_1}\right)$

$$f_X(x)=\frac{1}{2\pi\sigma_1}e^{-\frac{(x-\mu_1)^2}{2\sigma_1^2}}\int_{-\infty}^{+\infty}e^{-\frac{t^2}{2}}dt$$

$$=\frac{1}{\sqrt{2\pi}\,\sigma_1}e^{-\frac{(x-\mu_1)^2}{2\sigma_1^2}}\qquad -\infty<x<+\infty$$

$$X\sim N(\mu_1,\sigma_1^2)$$

The marginal distribution of continuous $r.v.$

$$(X,Y)\sim N(\mu_1,\mu_2,{\sigma_1}^2,{\sigma_2}^2,\rho)$$

$$f(x,y)=\frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}}e^{\frac{-1}{2(1-\rho^2)}\left[\frac{(x-\mu_1)^2}{\sigma_1^2}-\frac{2\rho(x-\mu_1)(y-\mu_2)}{\sigma_1\sigma_2}+\frac{(y-\mu_2)^2}{\sigma_2^2}\right]}$$

similarly,

$$f_Y(y)=\frac{1}{\sqrt{2\pi}\,\sigma_2}e^{-\frac{(y-\mu_2)^2}{2{\sigma_2}^2}} \qquad -\infty<y<+\infty$$

$$Y\sim N(\mu_2,{\sigma_2}^2)$$

The marginal distribution of continuous $r.v.$

$$f_X(x) = \frac{1}{\sqrt{2\pi}\,\sigma_1} e^{-\frac{(x-\mu_1)^2}{2\sigma_1^2}} \qquad f_Y(y) = \frac{1}{\sqrt{2\pi}\,\sigma_2} e^{-\frac{(y-\mu_2)^2}{2\sigma_2^2}}$$

Both marginal PDF of a 2-D normal distribution are 1-D normal distributions, and both are irrelevant to parameter ρ, Which means ρ describe the impact between r.v. X and Y.

Joint distribution $\rightleftarrows$ **Marginal distribution**

The marginal distribution of continuous $r.v.$

If $(X,Y) \sim N(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho)$, then $X \sim N(\mu_1, \sigma_1^2)$ and $Y \sim N(\mu_2, \sigma_2^2)$, and they don't depend on ρ.

- For bivariate normal distributions, they may have the same σ_1; σ_2; μ_1; μ_2 and different ρ, but their marginal probability density functions are the same one.
- For two-dimensional random variables, we have that the joint distribution can decide marginal distribution, but marginal distribution can't decide joint distribution.

Joint distribution $\rightarrow$ **Marginal distribution**

$\not\leftarrow$